

Lee Gene Ee

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EDUCATION

National University of Singapore

Aug 2022 - May 2026

Bachelor of Science in Business Analytics, minor in Statistics | CAP: 4.81 / 5.00

- Recipient of NUS Merit Scholarship and Canada-ASEAN Development Scholarship
- School of Computing Dean's List AY23/24

TECHNICAL SKILLS

Languages (ordered by proficiency): Python, SQL, Java, R, JavaScript, HTML/CSS

ML and Data: TensorFlow, PyTorch, CUDA, Scikit-learn, Spark, Hadoop, Airflow, dbt, Langchain, Ollama

Cloud & DevOps: AWS, Azure, Docker, Kubernetes

RELEVANT EXPERIENCE

Tencent – WeChat, Machine Learning Engineer Intern

Jan 2026 - Present

- Built and deployed end-to-end fraud detection systems using Graph SAGE-based GNNs with custom deep learning architecture in PyTorch, achieving **~0.91 F1-score** with separate precision and recall optimized models.
- Built **LLM-based agentic explainability framework** to interpret graph predictions and support investigations.
- Refactored graph inference pipeline by streaming data processing and redesigning memory-efficient batching to reduce RAM usage and cutting compute cost by **~62%** at production scale.

Tiktok, Machine Learning Engineer Intern

Jul 2025 - Dec 2025

- Built and deployed optimization algorithms with linear and constraint programming to improve task-labeler assignment, **reducing annotation cost per sample by ~26%** while sustaining quality.
- Implemented feature engineering workflows and data pipelines to streamline large-scale data acquisition processes.
- Deployed algorithms to production via ByteCloud in collaboration with Data Science and MLOps teams, enabling scalable integration with multimodal LLM training.

Govtech, Software Engineer Intern – Machine Learning

Jan 2025 - Jun 2025

- Developed and validated machine learning models, including regression and Graphical Neural Networks to quantify student learning outcomes, enabling data-driven scaling for **~300,000 students**.
- Built scalable data pipelines with AWS (S3, RDS) and Spark to process **10M+** rows daily, leveraging parallel computing services like Redshift to extract, clean, and analyze student learning data.
- Engineered a **full-stack LLM grading system** leveraging Azure OpenAI LLMs and Langchain; Handled end-to-end architectural design, prompt engineering and system evaluation, delivering projected cost savings of **~SGD 700,000** per school annually.

NUS Department of Information Systems, AI / ML Research Assistant

May 2024 - Aug 2024

- Built various time series architectures including LSTM and Transformer models to achieve a **63% baseline** F1 score in early-stage DDoS attack prediction.

PROJECTS

Utilizing Graph Neural Networks for Social Bot Detection ([Link](#))

- Submitted research paper to HICSS Conference under guidance of Professor Wang Qihong.
- Developed a novel Graph Neural Network (GNN) architecture that outperformed standard models like GAT and GCN for social bot detection

Knowledge-Guided Graph Based Approach for Legal RAG Application (Final Year Project)

- Built a hierarchical legal knowledge graph with IRAC-structured case law, supported by multi-strategy retrieval, interactive visual analytics, and provenance tracing using deterministic metrics and LLM-based evaluation (RAGAS, DeepEval), turning black-box LLM outputs into verifiable reasoning workflows.